

MONTHLY INTERNSHIP REPORT MAR - 2026

Student Name	Arpitha C
USN	4AI22CS008
Organization	Dhee Coding Lab
Internship Type (Paid/Unpaid)	Unpaid
Internship Duration	4 Months
Internship Name	Java Full Stack with Cloud and DevOps Essentials
Internship Mode	Offline
Report 2 Duration	01-03-2026 to 28-03-2026

WEEKLY WORK DESCRIPTION

Week 1: (02 Mar - 07 Mar 2026)

Focus: Java Inheritance & OOP Concepts, SQL Subqueries, and HTML Development

Work Description: During this week, the focus was on Java object-oriented concepts such as inheritance, method overriding, and variable shadowing. In SQL, subqueries and related concepts were practiced. In frontend, HTML tags and webpage development basics were covered.

Java:

- Variable shadowing and use of *this* keyword.
- Class diagrams and structure of classes.
- Inheritance and its types (single, multilevel, hierarchical, hybrid).
- Method overriding and its rules.
- Code reusability using inheritance.

Sql:

- Practice of subqueries.
- Types of subqueries.
- Use of ALL and ANY operators.
- Nested subqueries.
- Employee–manager relationship queries.

Frontend:

- Basic HTML tags: <h1>--<h6>, <p>,
, <hr>, <pre>.
- Text formatting tags: , <i>, <u>, , , <mark>.
- Anchor tag <a> and target attribute.
- Image and audio <audio> tags.
- Practical implementation of HTML pages.

Weekly Outcome: This week, I improved my understanding of Java inheritance, method overriding, and variable shadowing. I practiced SQL subqueries and learned how to use them for data retrieval. I also gained hands-on experience in HTML, which helped me understand how to structure and design basic web pages.

Week 2: (08 Mar – 14 Mar 2026)

Focus: Java Data Types & Casting, SQL Joins, and HTML Advanced Concepts

Work Description: During this week, the focus was on Java concepts such as data types, type casting, garbage collection, and object representation. In SQL, different types of joins and queries were practiced. In HTML, advanced topics like video, lists, tables, and forms were covered.

Java:

- Learned the concept of toString() method for object representation.
- Understood Java data types (primitive and non-primitive).
- Practiced variable assignment concepts.
- Learned garbage collection and its role in memory management.
- Understood type casting (primitive and non-primitive).

Sql:

- Practiced employee–manager relationship queries (Case 2).
- Learned different types of joins.
- Understood and executed inner join queries.
- Learned outer joins (left, right, full) with examples.
- Practiced queries on joins concept.
- Understood and executed self join queries.

Frontend:

- Learned <video> tag and its attributes (controls, autoplay, loop).
- Understood how to embed webpages.
- Learned semantic and non-semantic tags.
- Understood block-level and inline elements.
- Practiced list creation using , , <dl>.
- Learned table creation with <table>, <caption>, rowspan, colspan.
- Practiced creating forms and input fields.

Weekly Outcome: This week improved understanding of Java concepts, SQL joins, and HTML elements. Skills in programming, database queries, and webpage development were strengthened.

Week 3: (16 Mar – 21 Mar 2026)

Focus: Java Type Casting & Binding, SQL Functions, and HTML/CSS Development

Work Description: During this week, the focus was on Java concepts like type casting and method binding. In SQL, single row and string functions were practiced. In frontend, HTML forms and introduction to CSS were covered.

Java:

- Learned primitive type casting with examples.
- Understood upcasting and downcasting in inheritance.
- Practiced execution of type casting at runtime.
- Revised primitive and non-primitive casting concepts.
- Learned method binding (early binding and late binding).

Sql:

- Practiced joining multiple tables using joins.
- Learned single row functions (DUAL, CONCAT, UPPER, LOWER, INITCAP, REVERSE).
- Practiced string functions (SUBSTRING, REPLACE, INSTR).
- Learned formatting functions (LPAD, RPAD, TRIM, LTRIM, RTRIM).
- Practiced numerical functions (MOD, ROUND, TRUNC).
- Learned date-based SQL functions.

Frontend:

- Learned form input fields (text, date, number, file, radio, checkbox).
- Practiced form elements (dropdown, button, fieldset, legend).
- Executed HTML tags through practical sessions.
- Revised HTML concepts through exercises.
- Learned CSS types (inline, internal, external).
- Understood CSS selectors (element, class, id).

Weekly Outcome: This week improved understanding of Java type casting and method binding concepts. SQL skills were enhanced by practicing various functions and queries. Frontend knowledge was strengthened through HTML form handling and basic CSS styling.

Week 4: (23 Mar – 28 Mar 2026)

Focus: Java Polymorphism & Static Concepts, SQL Queries & DDL, and CSS Styling

Work Description: During this week, the focus was on Java concepts like polymorphism, static keyword, and method binding. In SQL, advanced queries, functions, subqueries, and DDL commands were covered. In frontend, CSS selectors and styling properties were practiced.

Java:

- Learned method binding (early and late binding).
- Understood polymorphism (compile-time and runtime).
- Practiced method overloading and overriding concepts.
- Learned static keyword, static variables, static block, static methods, and static import.
- Attempted Java test for concept evaluation.

Sql:

- Learned date functions (SYSDATE, CURRENT_DATE, LAST_DAY, ADD_MONTHS, MONTHS_BETWEEN, TO_DATE, TO_CHAR).
- Practiced NVL and NVL2 functions.
- Understood pseudo columns (ROWID, ROWNUM).
- Practiced correlated subqueries.
- Learned queries to find nth max and min salary.
- Understood set operators (UNION, UNION ALL, INTERSECT, MINUS).
- Learned DDL commands (CREATE, RENAME, ALTER, TRUNCATE).

Frontend:

- Learned CSS selectors (simple, grouping, universal, attribute, nth-child, pseudo-class).
- Understood color properties (name, RGB, HEX, HSL).
- Learned background properties (color, image, repeat, position, size).
- Practiced box model concepts.
- Learned border and border-radius properties.
- Understood outline property and styling.

Weekly Outcome: This week improved understanding of Java polymorphism, static concepts, and method binding. SQL skills were enhanced with advanced queries, functions, and DDL commands. CSS knowledge was strengthened through selectors and styling properties.

Snapshots:

2026-03-13	The session covered type casting in Java, focusing on non-primitive casting and its usage. In SQL, the natural join was explained and queries related to different types of joins were practiced. In HTML, tables were discussed along with tags and attributes such as <table>, <caption>, rowspan, and colspan.	 
2026-03-12	The session covered garbage collection in Java with examples and an introduction to type casting. In SQL, outer joins and their types such as left join, right join, and full join were explained along with related queries. In HTML, list tags were discussed and demonstrated.	 
2026-03-11	The session covered primitive and non-primitive variables in Java and the concept of variable assignment. In SQL, the inner join concept was explained and related queries were executed to understand how data from multiple tables can be combined. In HTML, different types of lists were discussed including unordered lists, ordered lists, description lists, and nested lists.	 
2026-03-10	The session covered Java data types and their two main categories along with explanations. In SQL, the concept of joins and the five types of joins were discussed, and the Cartesian join was explained with related queries. In HTML, semantic and non-semantic tags were explained along with block-level and inline elements.	 
2026-03-09	The session covered the concept of toString() in Java with examples to understand how objects can be represented as strings. In SQL, the employee-manager relationship (Case 2) was explained, where employee details were identified using manager details, along with related queries. In HTML, the <video> tag and its additional attributes were discussed, and methods to add other webpages into an existing webpage were explained.	 
2026-03-07	A practical session was conducted on HTML where various tags were implemented and executed to understand their usage in webpage development. In SQL, the concept of the employee-manager relationship (Case 1) was explained to identify reporting manager details using employee table data. Related queries were executed to understand the relationship between employees and their managers.	 
2026-03-06	The session covered HTML concepts including the anchor tag and its target attribute, image tag, and audio tag used in webpage development. In SQL, nested subqueries were explained and related queries were executed to understand how subqueries can be used within another subquery.	 
2026-03-05	The session covered method overriding in Java, its rules, and examples to understand how a subclass can redefine a method of its parent class. In SQL, types of subqueries were explained along with the use of ALL and ANY operators. Related queries were executed to understand their usage in filtering data.	 
2026-03-04	Learned the concept of Java inheritance and practiced creating parent and child classes using the extends keyword. Also practiced SQL queries for retrieving data and worked on HTML basics to understand webpage structure.	 
2026-03-03	Learned about inheritance in Java and how one class can acquire properties and methods of another class. Also practiced using different HTML tags for structuring web pages	 
2026-03-02	Revised the this keyword in Java and practiced its usage in programs. Also continued learning and implementing different HTML tags in frontend development.	 

2026-03-28	A test was conducted in Java to evaluate understanding of the concepts covered. In SQL, Data Definition Language commands such as ALTER and TRUNCATE were explained along with related queries. In HTML/CSS, the outline property was discussed and practical execution of concepts was carried out.	 
2026-03-27	The session covered static block, static methods, and static import in Java. In SQL, set operators were revised and Data Definition Language (DDL) commands such as CREATE and RENAME were discussed. In HTML/CSS, box model concepts and border properties including border and border-radius were explained.	 
2026-03-26	The session covered the static keyword in Java, including static variables with examples. In SQL, continuation of correlated subqueries was discussed to find nth maximum and nth minimum salary, and set operators such as UNION, UNION ALL, INTERSECT, and MINUS were explained. In CSS, background properties were covered including background color, background image, background repeat, background attachment, background position, and background size.	 
2026-03-25	The session covered polymorphism in Java, including compile-time and runtime polymorphism. In SQL, pseudo columns such as ROWID and ROWNUM were explained along with related queries, and correlated subqueries were also discussed. In HTML/CSS, CSS properties related to color were covered, including different ways to define colors such as built-in names, RGB, hexadecimal, and HSL.	 
2026-03-24	The session covered examples of method binding in Java, including early binding and late binding to understand compile-time and runtime behavior. In SQL, date-based single row functions such as LAST_DAY, SYSDATE, CURRENT_DATE, TO_DATE, MONTHS_BETWEEN, ADD_MONTHS, and TO_CHAR were explained along with NVL and NVL2 functions, and related queries were executed. In HTML/CSS, different types of CSS selectors were discussed including simple selectors, grouping, universal, attribute selectors, nth-child, and pseudo-class selectors.	 
2026-03-23	The session covered method binding in Java, including early binding and late binding concepts. In SQL, numerical single row functions such as MOD, ROUND, and TRUNC, along with date-based single row functions, were explained and practiced through queries. In HTML/CSS, an introduction to CSS was given, including types of CSS (inline, internal, external) and different types of CSS selectors.	 
2026-03-20	The session focused on revision of type casting in Java to reinforce understanding of primitive and non-primitive conversions. In SQL, functions such as LPAD, RPAD, TRIM, RTRIM, and LTRIM were practiced through executed queries. In HTML, a practical session was conducted where various HTML tags were implemented by solving topic-based questions.	 
2026-03-18	The session focused on execution examples of upcasting and downcasting in Java to understand runtime behavior. In SQL, string functions such as LPAD, RPAD, TRIM, RTRIM, and LTRIM were explained along with related queries. In HTML, a practical session was conducted where various HTML tags were executed to understand their implementation.	 
2026-03-17	The session included examples of upcasting and downcasting in Java to understand type conversion in inheritance. In SQL, string functions such as SUBSTRING, REPLACE, and INSTR were explained along with related queries. In HTML, form elements like dropdown list, button, fieldset, and legend were discussed. A practice session was also conducted to revise previously covered topics.	 
2026-03-16	The session covered the concepts of upcasting and downcasting in Java and how they are used in inheritance. In SQL, single row functions were discussed along with examples using DUAL, CONCAT, UPPER, LOWER, INITCAP, and REVERSE. In HTML, different input fields in forms were explained such as text field, date, number, file upload, radio button, and checkbox.	 
2026-03-15	The session covered the concept of primitive type casting in Java with examples to understand how values can be converted between primitive data types. In SQL, joining multiple tables was explained and related queries were executed to retrieve combined data from different tables.	 
2026-03-14	The session covered type casting in Java with focus on primitive casting and its types. In SQL, the concept of self join was explained and related queries were executed. In HTML, forms were discussed along with different types of form fields such as text fields and the use of the id attribute.	 